Environmental Science & Technology

pubs.acs.org/est

Article

Open Access

This article is licensed under CC-BY-NC-ND 4.0

Occurrence of Major Perfluorinated Alkylate Substances in Human Blood and Target Organs

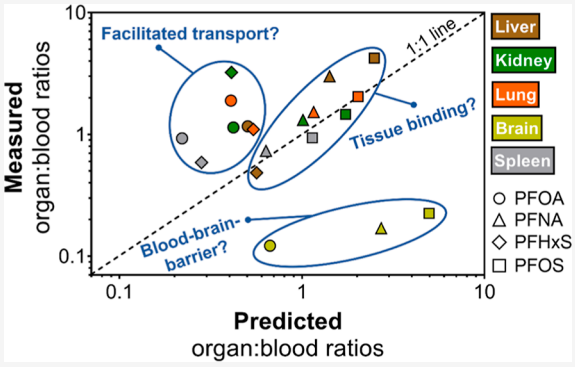
Flemming Nielsen, Fabian C. Fischer, Peter M. Leth, and Philippe Grandjean*

Cite This: *Environ. Sci. Technol.* 2024, 58, 143−149

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Human exposure to perfluorinated alkylate substances (PFASs) is usually assessed from the concentrations in serum or plasma, assuming one-compartment toxicokinetics. To characterize body distributions of major PFASs, we obtained and extracted tissue samples from 19 forensic autopsies of healthy adult subjects who had died suddenly and were not known to have elevated levels of PFAS exposure. As target organs of toxicological importance, we selected the liver, kidneys, lungs, spleen, and brain, as well as whole blood. Samples weighing about 0.1 g were analyzed by liquid chromatography coupled to triple mass spectrometers. Minor variations in PFAS concentrations were found between the kidney cortex and medulla and between lung lobes. Organ concentrations of perfluorooctanoic sulfonate (PFOS) and perfluorononanoate (PFNA) correlated well with blood concentrations, while perfluorooctanoate (PFOA) and perfluorohexanoic sulfonate (PFHxS) showed more variable associations. Likewise, the liver concentrations correlated well with those of other organs. Calculations of relative distributions were carried out to assess the interdependence of organ retentions. Equilibrium model predictions largely explained the observed PFAS distributions, except for the brain. Although the samples were small and affected by a possible lack of homogeneity, these findings support the use of blood-PFAS concentrations as a measure of PFAS exposure, with the liver possibly acting as the main organ of retention.



KEYWORDS: *blood, brain, kidney, liver, lung, organ retention, PFAS*

■ INTRODUCTION

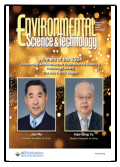
Per- and polyfluorinated alkylate substances (PFASs) have caused global contamination, and thus human exposure and risk of toxicity have become a public health concern. PFASs can cause liver toxicity, e.g., as indicated by elevated lipid concentrations and liver enzymes in serum as well as kidney toxicity.[1] In addition, PFAS immunotoxicity is well documented[2] and may well involve the spleen. An increased risk of infection may particularly involve the lungs.[3,4] Also, the brain may be a target organ for PFASs.[5] In contrast to many other organic pollutants, PFASs are not retained in fat tissue, but organ distributions of PFASs are poorly known. Adverse effects have been found in humans at lower and lower PFAS concentrations measured in blood.[6] Likewise, PFAS half-lives have been calculated on the basis of serial blood concentrations, which may or may not reflect levels in body organs.[7]

Unfortunately, little information is available on this important aspect of PFAS toxicology and the factors that may affect the relative concentrations in the target organs. Such information will likely be useful for further improving human physiologically based toxicokinetic (PBTK) models.[8,9]

Past studies have examined only small numbers of tissue samples from deceased adults, viz., five different tissues from 20 subjects with a stable residence in Tarragona, Spain, who died from various causes,[10] and surgical materials from nine cancer patients in France.[11] These autopsy samples may not be representative of the general population, but they provide some insights into relative organ distributions.

PBTK modeling has been used to describe the body distribution of PFASs in mammals,[12] but their applicability to human exposure is currently limited due to missing toxicokinetic parameters, such as permeabilities, transport affinities, and elimination constants.[13] As a simplified alternative, equilibrium distribution modeling (EDM) can predict the distribution of PFASs based on the protein, lipid, and water content of various human organs.[14] Comparing EDM simulations to PFAS concentrations observed in human organs could offer valuable insights into whether organ binding primarily drives the overall organ distribution or if kinetic processes need to be accounted for in more complex PBTK

Received: August 10, 2023
Revised: December 1, 2023
Accepted: December 1, 2023
Published: December 28, 2023



Downloaded via NATL TECHL UNIV OF ATHENS on March 31, 2025 at 16:55:20 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

ACS Publications

© 2023 The Authors. Published by American Chemical Society

models (e.g., enterohepatic recirculation and renal reabsorption).

To explore relative PFAS concentrations in major organs, we obtained anonymized tissue samples from forensic autopsies of unexpected deaths in Denmark. We assessed the most prevalent PFASs in the small tissue samples using a preparation procedure involving salting out the PFASs from the homogenized tissue into acetonitrile prior to online solid-phase extraction (SPE), chromatographic separation, and quantitation by triple quadrupole mass spectrometry (LC−MS/MS). We then determined the capabilities of EDM to offer a coherent explanation for the relative organ concentrations observed.

■ MATERIALS AND METHODS

Sample Collection. Tissue samples were obtained in 2020−2021 from 19 medicolegal autopsies of sudden deaths that required forensic examination. All subjects resided in the region of Southern Denmark. We sampled major tissues of particular relevance to the PFAS toxicity. Thus, peripheral lung tissue was obtained from all five lobes. Kidney tissue was from both the cortex and medulla on the right side. In addition, tissue was sampled from the liver, spleen, and motor cortex of the brain. Finally, whole blood was sampled from the common iliac vein. Causes of death varied, but all subjects died suddenly, and none was hospitalized or under treatment for serious disease. In general, autopsies and sample collection took place within the first 24 h after the time of death. The samples were coded, and the general information made available included age and sex. Samples were frozen immediately after autopsy and stored at −80 °C until analysis. The project was reviewed and approved by the ethical review committee serving the region of southern Denmark (project number 20200143).

Sample Preparation. Tissue samples were analyzed after thawing overnight at 5 °C. Biopsies were drip-dried on a lint-free paper tissue and accurately weighed into a tared 2 mL polypropylene microtube (SARSTEDT, Nümbrecht, Germany). The samples ranged in weight from 0.033 to 0.162 g, with an average weight of 0.143 g. Whole blood samples were pipetted with a fixed volume of 150 μL. To all tissue samples, we added 250 μL of 0.5% formic acid in water, 30 μL of 20 ng/mL isotope-labeled PFAS analogues (as internal standards), and three 5 mm steel balls. The steel balls had, prior to use, been thoroughly cleaned by sonication in 0.1% Liquinox detergent solution, water, and methanol, respectively. Homogenization was performed on a Retsch Mixer Mill 200 at 25 Hz for 3 × 2.5 min (though 4 × 2.5 min for lung tissue). Hereafter, 75 mg of RoQ salts (60 mg magnesium sulfate and 15 mg sodium chloride) (Phenomenex, Torrance, CA) was added, in order to remove excess water from the sample and induce phase separation between the water and the acetonitrile layer. The samples were vortex-mixed for 1 min and centrifuged at 21.000 *g* for 15 min, after which 180 μL of the supernatant was transferred to a Whatman Mini-UniPrep polypropylene syringeless filter vial (Whatman Inc., Florham Park, NJ), followed by addition of 450 μL of 0.1 M formic acid. The plunger unit containing a 0.45 μm nylon filter and a vial lid with septum was then compressed into the vial housing, filtering particulates out of the sample. The subsequent analytical procedure was identical to the method the laboratory utilizes for serum and whole blood samples.[15,16] A sample volume of 400 μL was injected into the online SPE column.

Due to the small size of most tissue samples, they were not dissected further and therefore weighed directly into the microtube as a single piece. Slightly larger-sized samples were dissected to obtain a specimen of approximately 0.10 g before being weighed into the vial. Likewise, the partially or fully coagulated blood samples did not allow separation into serum and blood cells and were analyzed as whole blood. Due to possible additional changes from hemolysis or simple gravity after the time of death, we also assessed the hemoglobin concentration of the samples. One sample from the right lung upper lobe and five samples from the brain were too small (less than 0.05 g) to be appropriately analyzed and were therefore considered as missing.

PFAS Analysis. The extraction procedure for the tissue samples was optimized using the liver, kidney, and lung tissue samples from sheep and pigs, provided by the Biomedical Laboratory that serves as the central animal facility at the University of Southern Denmark and Odense University Hospital. The tissue samples were fortified to contain 0.400 ng/g with a PFAS solution containing perfluorobutanoate (PFBA), perfluorohexanoic sulfonate (PFHxS), perfluorooctanoate (PFOA), perfluorooctanoic sulfonate (PFOS), perfluorononaoate (PFNA), perfluorodecanoate (PFDA), and perfluoroundecanoate (PFUnDA) (Wellington Laboratories, Inc., Guelph, Ontario, Canada) and homogenized as described above. The relative PFAS recovery in whole blood was assessed by analyzing in-house quality control (QC) material made from a pool of whole blood collected from healthy anonymous donors according to local guidelines and regulations at Odense University Hospital, Denmark. The mean relative recovery for the selected PFASs ranged from 86.3 to 111.0% (see Supporting Information). The limits of detection (LODs) were 0.03 ng/g of tissue wet weight, except for PFHxS in brain cortex tissue (0.06 ng/g). All batches of samples included serum quality control samples from the German External Quality Assessment Program (G-EQUAS) as well as NIST SRM 1957 (National Institute of Standards and Technology, Gaithersburg, MD), which were extracted the same way as the tissue samples, and the results of these analyses were well within the acceptance ranges. The accuracy of the control samples calculated as bias was <9% for all compounds.

The extracts were analyzed using a Thermo Fisher Scientific EQuan MAX UHPLC system (consisting of an Ultimate 3000 RS pump, an Ultimate 3000 LPG pump, an Ultimate 3000 RS column compartment, and a CTC PAL autosampler), connected to a TSQ Quantiva triple quadrupole mass spectrometer (Thermo Scientific, San Jose, CA). The online SPE and the gradient elution of the analytical column were performed by the EQuan MAX chromatographic system. All instruments were controlled, and all PFAS data were calculated using Xcalibur software (version 4.5) (Thermo Scientific, San Jose, CA). The injected sample was during the online SPE procedure trapped on an ACE C18 2.1 mm × 10 mm column and washed for 6 min (transfer time) by a mobile phase consisting of 0.1 M formic acid/methanol (95:5) at a flow rate of 300 μL/min. The PFASs were then, after column switching, transferred to a HALO 2.1 mm × 100 mm (2.7 μm, 90 Å) PFAS column for chromatographic separation (Advanced Material Technology, Wilmington, DE). The analytes were eluted from the analytical column using a binary gradient with (98:2) 0.1% acetic acid: methanol (mobile phase A), and (98:2) methanol with 0.1% acetic acid: 2 mM ammonium acetate (mobile phase B) in the following elution profile: 0−6

https://doi.org/10.1021/acs.est.3c06499
Environ. Sci. Technol. 2024, 58, 143−149

min: 5% B; 6−7 min: 5−30% B; 7−18 min: 30−100% B; 18−21 min: 100% B, 21−21.2 min: 100−5% B; 21.2−25 min: 5% B. The flow rate was 300 μL/min. A HALO 4.6 mm × 50 mm (2.7 μm, 160 Å) PFAS Delay column was inserted between the pumps and the injector to retain eventual system- and mobile phase-related PFAS interferences. The TSQ Quantiva was operated with heated electrospray ionization (H-ESI) in negative mode using the following settings: spray voltage: 2500 V; vaporizer temperature: 300 °C; sheath gas: 50 (Arb); auxiliary gas: 10 (Arb); sweep gas: 1 (Arb); ion transfer tube temperature: 325 °C.

All stock PFAS reference substances (PFAC-24PAR) as well as the mass-labeled PFAS analogues (MPFAC-24ES) (Internal Standards) were purchased as mixtures at concentrations of 1 μg/mL in methanol from Wellington Laboratories Inc. (Guelph, Ontario, Canada). These solutions contain a mixture of 24 PFASs (for further details, see the Web site of Wellington Laboratories, www.well-labs.com). Working solutions were also prepared in methanol. Methanol, formic acid, and acetic acid were purchased from VWR International (Radnor, PA, USA).

Hemoglobin Analysis. The whole blood samples were analyzed for hemoglobin on a Konelab 20 Clinical Chemistry analyzer (Thermo, Vantaa, Finland). A volume of 20 μL whole blood was added into a 5.0 mL Drabkińs solution in a 15 mL reagent tube (Sarstedt, Nümburg, Germany). The sample was briefly whirly mixed and left to stand for 5 min prior to analysis. Calibrator and control samples were provided from Eurotrol BV (BS Ede, Netherlands). The accuracy of the control samples calculated as bias was <2%.

Statistical Analysis and Equilibrium Distribution Modeling. Given the small number of samples, non-quantifiable results were assumed to be half of the detection limit. Deviations from normal distributions were minor, and data are therefore presented as means and standard deviations. Standard partial correlations were calculated to allow adjustment for covariables, including blood-hemoglobin, sex, and age. PFAS concentrations in organ samples were compared to those in whole blood and liver as the main compartments of interest.

An EDM[14,17] was adapted to determine if the organ/blood and organ/liver concentration ratios observed matched predictions based on the composition of the organs and corresponding partition coefficients. The tissues analyzed were included as model compartments, and the organs were assumed to not contain blood. The EDM assumes equilibrium between the organs, i.e., omitting membrane permeability, reuptake, elimination, and transformation. Organ/blood and organ/liver concentration ratios were predicted solely based on the binding affinity of the PFASs to the individual components of the organ, i.e., their protein and lipid contents. We based the calculation on the assumption that the organs were composed of albumin (Alb), fatty acid-binding proteins (FABP), structural proteins (SP), membrane phospholipids (ML), and nonsorptive components (water, storage lipids, and sugars). The average content of these components in human blood, lung, liver, spleen, kidney, and brain tissue was reported in Allendorf[14,17] and slightly updated as described in our recent EDM paper.[17] Partition coefficients (the concentration ratio between two phases at equilibrium) were used as model parameters to quantify the binding of PFASs to these organ constituents. The relative protein and lipid composition of organs defined the predicted organ/blood and organ/liver concentration ratios, as previously described.[18] Alb-water, FABP-water, SP-water, and ML-water partition coefficients for the study of PFASs were gathered from the literature.[17−26] The predicted organ/blood and organ/liver concentration ratios were then compared to the organ concentrations observed.

RESULTS AND DISCUSSION

The organ tissue samples obtained from the medicolegal autopsies originated from subjects aged between 29 and 84 years; 5 were women and 14 were men. The small sizes of the tissue samples generally did not allow further dissection and were mostly taken as a whole for PFAS analysis. Likewise, the blood samples were partially or fully coagulated and did not allow separation into serum and erythrocytes; these whole blood samples were therefore analyzed without any pretreatment, despite somewhat differing compositions. Four PFASs (PFHxS, PFOA, PFOS, and PFNA) were detectable in the liver, kidney, and lungs, while the spleen and brain had much fewer detections (Table 1). The results represent the sum of linear and branched substances. The highest concentrations found originated from one of the oldest males (>80 years) in the study.

Table 1. Mean and Standard Deviation of PFAS Concentrations (ng/g Wet Weight) and Numbers with Detectable Concentrations in Samples from 19 Medicolegal Autopsies

tissue	PFHxS	PFOS	PFOA	PFNA
liver	0.074 (0.052)	6.277 (5.497)	0.324 (0.159)	0.336 (0.219)
number detected	15	19	19	19
kidneys	0.494 (0.394)	2.174 (2.062)	0.317 (0.173)	0.147 (0.097)
number detected	19	19	19	18
lungs	0.168 (0.148)	3.031 (2.427)	0.526 (0.214)	0.171 (0.104)
number detected	18	18	19	18
spleen	0.090 (0.079)	1.392 (1.332)	0.258 (0.153)	0.082 (0.097)
number detected	16	19	17	8
brain cortex	ND[a]	0.334 (0.532)	0.034 (0.028)	0.019 (0.016)
number detected	0	8	7	1
blood	0.153 (0.128)	1.486 (1.569)	0.278 (0.180)	0.112 (0.090)
number detected	18	19	19	16

[a]PFHxS in all samples from the brain cortex was below LOD (0.06 ng/g). The LODs for PFHxS, PFOS, PFOA, and PFNA in all other tissues were 0.03 ng/g.

While PFAS concentrations varied somewhat between the samples from the five different lung lobes, averages correlated well with levels in the individual lobes, i.e., r = 0.72−0.92 for PFHxS, 0.72−0.91 for PFOS, 0.70−0.87 for PFOA (except for the upper right lobe that showed r = 0.116), and 0.57−0.80 for PFNA (although lower lobes on the right and left showed r = 0.200 and 0.362). Overall, PFHxS, PFOS, and PFOA concentrations in the right upper lung lobe correlated less well with those in the other four lobes, although the opposite

https://doi.org/10.1021/acs.est.3c06499
Environ. Sci. Technol. 2024, 58, 143−149

tendency was seen for PFNA. Nonetheless, the overall average PFAS concentrations in the lung lobes were used in further calculations. Furthermore, PFAS concentrations in the kidney cortex correlated well with those in the kidney medulla (R = 0.84−0.96), thus justifying using the average kidney concentrations for additional analyses.

The liver contained much more PFOS than was found in other organs, and this pattern was also apparent for PFNA (Table 1). PFHxS was the only PFAS that showed higher concentrations in the kidney than in the liver, while PFOA was higher in the lungs than in the liver. Concentrations in the brain and spleen were comparatively low. In fact, only PFOS was detectable in the majority of brain samples, while other PFASs were mostly below the detection limit.

For comparison, a study carried out in Spain[10] showed PFAS concentrations in human organs that were much higher than those determined in the present study, perhaps in part because the Spanish samples were collected about 10 years earlier, when exposure to the legacy PFASs tested here was much higher. Still, the results confirm that PFOS mainly accumulates in the liver, kidney, and lungs and that concentrations in the brain tend to be very low. High PFOS concentrations in the liver were also observed in an earlier study of 12 liver samples from autopsies in France.[27] The highest concentrations in the lungs have been reported for PFBA,[10] but this substance could not be reliably detected in the present study.

Interorgan correlations were then calculated for the tissues where PFASs were detected in all samples. Given the differing compositions of the blood, the partial correlation analysis of PFAS concentrations in the blood included the hemoglobin concentration as an independent variable along with sex and age (Table 2). In these calculations, average PFAS concentrations in the lung and kidney were used.

Table 2. Partial Correlation Coefficients for PFAS Concentrations in Whole Blood and in Organ Tissues, with Age, Sex, and Blood Hemoglobin as Independent Predictors (N = 19)[a]

tissue	blood			
	PFHxS	PFOS	PFOA	PFNA
liver	0.81	0.75	0.56	0.70
kidneys	0.54	0.85	0.69	0.77
lungs	0.79	0.94	0.79	0.90
spleen	0.73	0.88	0.56	0.69
brain cortex	ND[b]	0.90	0.67	0.70

[a]All coefficients are statistically significant ($p < 0.05$). [b]ND, not detected in the brain cortex.

In general, tissue concentrations correlated fairly well with those in the blood (Table 2). One exception was PFHxS that showed weaker correlations, especially for the kidney concentrations. PFOA correlated less well in the liver and spleen. Concentrations in the brain were low but did suggest higher retention levels at higher blood concentrations, except for PFHxS where measurements below the detection limit did not allow a reliable calculation. In regard to the liver concentrations, kidney PFHxS showed weak correlations, but correlations were fairly similar to those for blood (Table 3).

An important concern in studies like this one is that the tissue samples may not be fully representative of the respective organs. Some of the observed variations between tissue concentrations may be ascribed to uneven composition of the specific tissue sample, not only for the lung and kidney. Thus, small sample sizes (e.g., 1 g in the Spanish study,[10] and 0.1 g in the present study) may not include equal amounts of specific organ tissue, as connective tissue, fat, and blood vessels may be present in variable amounts. Likewise, the tissues removed by surgery of cancer patients in a previous study[11] may not represent premorbid PFAS concentrations.

Table 3. Partial Correlation Coefficients for PFAS Concentrations in Liver and Other Tissues, with Age and Sex as Independent Predictors (N = 19)

tissue	PFHxS	PFOS	PFOA	PFNA
kidneys	0.36	0.80[a]	0.52[a]	0.83[a]
lungs	0.66[a]	0.82[a]	0.61[a]	0.61[a]
spleen	0.50[a]	0.71[a]	0.45	0.68[a]
brain cortex	ND	0.85[a]	0.61[a]	0.52

[a]$p < 0.05$; ND, not detected in brain cortex.

Post-mortem processes change the blood cell distribution and the blood composition, thus also potentially affecting the concentration or location of bloodborne PFASs in the tissues.[28] Such a redistribution is poorly understood. In lung tissue, for example, post-mortem sedimentation of bronchial mucus, edema fluid, and blood likely occurs, often referred to as internal lividity.[29] This process may potentially cause changes in PFAS concentrations after death within the tissue, while post-mortem metabolism of PFAS precursors by microbes would appear unlikely during the short interval before autopsy.

Given that PFAS intake mostly occurs via the gut, the absorbed amount must first pass through the liver. PFAS toxicity includes elevated serum-lipid concentrations and other abnormalities that point to the liver as a major target organ.[1] Accordingly, especially, PFOS and PFOA showed average organ concentrations that correlated well with levels in both the blood and liver. Because urinary excretion is a potentially important elimination pathway, retention in the kidneys is also highly relevant, and kidney cancer has been shown to be a possible outcome, e.g., after PFOA exposure.[30] The spleen is of possible relevance due to the immunotoxic effects of PFAS exposure.[31] The brain has also been suggested as a target for PFASs and linked to some types of neurotoxicity.[5] Finally, the lungs are relevant as an entryway of inhaled PFASs and as a possible target for some PFAS-associated pulmonary infections.[3]

PFAS concentrations observed in these organs were used in an attempt to model the relative concentrations, especially with regard to retention in the blood and in the liver (Figures 1 and 2). The high retention of PFHxS in the kidney and the relatively poor correlation with blood concentrations suggest that PFHxS may be less efficiently eliminated in the urine and/or that tubular renal reabsorption is particularly pronounced as compared to other PFASs. On the other hand, retention of PFHxS, albeit at a lower level, seems to be reasonably predicted from blood and liver concentrations.

The organ/blood concentration ratios tended to be slightly higher than expected by the EDM for kidneys, lungs, and liver, whereas brain/blood partitioning was overestimated by a factor of ∼10 (Figure 1). Despite the small sample sizes and possible differences in composition, most organ/blood predictions were accurate within a factor of 3. Lung/blood and liver/blood distributions were particularly well predicted. Relative PFOA

https://doi.org/10.1021/acs.est.3c06499
Environ. Sci. Technol. 2024, 58, 143−149

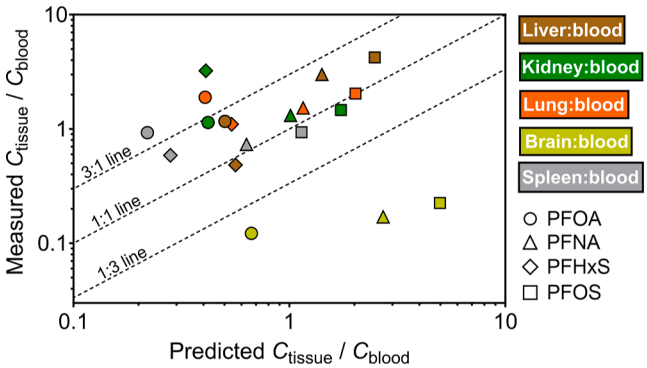


Figure 1. Predicted and measured tissue/blood concentration ratios of the four study PFASs.

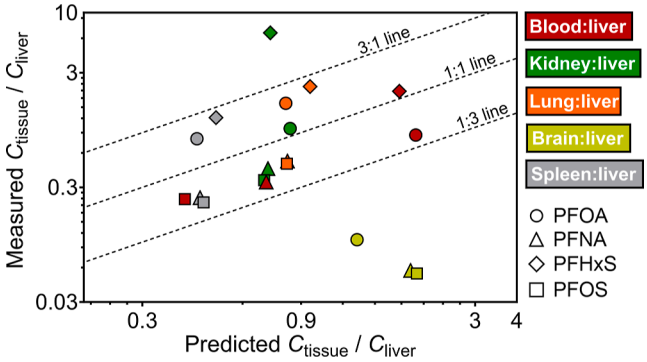


Figure 2. Predicted and measured tissue/liver concentration ratios of the four study PFASs.

enrichment in all organs but the brain surpassed EDM predictions, with the kidneys exhibiting an extent of PFHxS enrichment that was higher than anticipated by the model. In contrast, predictions matched the observed values well for PFNA and PFOS. Lung and liver were correlated for all PFASs determined and matched model expectations (Figure 2). Kidney/liver and spleen/liver also correlated for three of the four PFASs (Figure 2). As was already demonstrated for organ/blood ratios (Figure 1), brain/liver concentration ratios were overpredicted by the model by a factor of ∼10 (Figure 2).

Because our EDM does not account for poorly known and complex toxicokinetic processes like permeability, active membrane transporters, and elimination, especially, brain retention was strongly overpredicted by the model (as it does not account for the blood−brain barrier). PFHxS was observed at higher average concentrations in the kidney, as predicted by the model, which could be the result of facilitated transport by OATs or OATPs and subsequent tubular renal reabsorption.[32] While PFHxS is predominantly bound to proteins, especially albumin,[14] PFOS and PFNA show high sorption affinities to membrane phospholipids.[33] For these relatively hydrophobic PFASs, transporters are expected to be less relevant to organ distribution as compared to the tissue retention triggered by the presence of membrane phospholipids. The EDM results support this hypothesis, as, particularly, PFOS and PFNA kidney/liver distributions were accurately predicted, viz., two organs with particularly high membrane lipid content. Liver/blood ratios of PFOS and PFNA were underestimated by the model by a factor of ∼2, perhaps as a result of enterohepatic recirculation of the chemicals from the gut back to the liver.[34] However, coagulation of the blood samples might have resulted in reduced blood concentrations, which may have generated a higher organ/blood concentration than predicted by the EDM.

Thus, reasonable agreement was obtained between actual retention levels and the EDM predictions, thereby suggesting that organ distribution of PFASs may be primarily driven by tissue binding, especially for longer C-chain PFASs like PFOS and PFNA that exhibit high binding affinities to membrane lipids and high permeabilities (i.e., permeability is less rate-limiting).[20] For less-permeable PFASs that show weaker binding to tissue proteins and lipids,[19,20] active transport and reabsorption processes are expected to be more important, which may explain why the EDM underpredicts their organ enrichment. While active transport processes or barriers will generally affect tissue concentrations, our results confirm that physiological EDM is a useful, albeit simplistic, method to predict organ distribution in the absence of detailed data on membrane transporter affinities and permeabilities that would be required for reliable PBTK modeling. Furthermore, the existence of correlations (Table 2) between PFAS concentrations in blood and major organs provides support for the use of blood levels as exposure biomarkers in risk assessment.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.3c06499.

Laboratory quality data on recovery of PFASs fortified in tissues and blood (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Philippe Grandjean − *Department of Environmental Medicine, University of Southern Denmark, Odense 5230, Denmark; Department of Biomedical and Pharmaceutical Sciences, University of Rhode Island, Kingston, Rhode Island 02881, United States;* orcid.org/0000-0003-4046-9658; Phone: +45 6550 3769; Email: pgrand@sdu.dk

Authors

Flemming Nielsen − *Department of Environmental Medicine, University of Southern Denmark, Odense 5230, Denmark*

Fabian C. Fischer − *John A. Paulson School of Engineering and Applied Sciences, Harvard University, Cambridge, Massachusetts 02138, United States; Department of Biomedical and Pharmaceutical Sciences, University of Rhode Island, Kingston, Rhode Island 02881, United States*

Peter M. Leth − *Department of Forensic Medicine, University of Southern Denmark, Odense 5230, Denmark*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.est.3c06499

Author Contributions

All authors contributed to the study conception and design. Material selection was performed by Peter Leth and sample analysis by Flemming Nielsen. Statistical analysis was performed by Philippe Grandjean and organ distribution calculations by Fabian Fischer. The first draft of the manuscript was written by Philippe Grandjean, and all authors commented on the draft versions of the manuscript. All the authors read and approved the final manuscript.

https://doi.org/10.1021/acs.est.3c06499
Environ. Sci. Technol. 2024, 58, 143−149

Notes

The authors declare the following competing financial interest(s): P.G. has provided paid expert assistance in legal cases involving PFAS-exposed populations. All other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

■ ACKNOWLEDGMENTS

P.G. and F.C.F. were supported by the National Institute of Environmental Health Sciences (P42ES027706). The expert assistance of Vibeke K. Petersen in PFAS analyses is gratefully acknowledged.

■ REFERENCES

(1) Agency for Toxic Substances and Disease Registry (ATSDR). Toxicological profile for perfluoroalkyls. *U.S. Department of Health and Human Services PHS*; Agency for Toxic Substances and Disease Registry: Atlanta, GA, 2021.

(2) Budtz-Jorgensen, E.; Grandjean, P. Benchmark dose calculations for PFAS exposure based on two data sets on immunotoxic effects. *Environ. Health* **2023**, *22* (1), 40.

(3) Grandjean, P.; Timmermann, C. A. G.; Kruse, M.; Nielsen, F.; Vinholt, P. J.; Boding, L.; Heilmann, C.; Mølbak, K. Severity of COVID-19 at elevated exposure to perfluorinated alkylates. *PLoS One* **2020**, *15* (12), No. e0244815.

(4) Dalsager, L.; Christensen, N.; Halekoh, U.; Timmermann, C. A. G.; Nielsen, F.; Kyhl, H. B.; Husby, S.; Grandjean, P.; Jensen, T. K.; Andersen, H. R. Exposure to perfluoroalkyl substances during fetal life and hospitalization for infectious disease in childhood: A study among 1,503 children from the Odense Child Cohort. *Environ. Int.* **2021**, *149*, 106395.

(5) Goodman, C. V.; Till, C.; Green, R.; El-Sabbagh, J.; Arbuckle, T. E.; Hornung, R.; Lanphear, B.; Seguin, J. R.; Booij, L.; Fisher, M.; et al. Prenatal exposure to legacy PFAS and neurodevelopment in preschool-aged Canadian children: The MIREC cohort. *Neurotoxicol. Teratol.* **2023**, *98*, 107181.

(6) U.S. Environmental Protection Agency. Technical Fact Sheet: Drinking Water Health Advisories for Four PFAS (PFOA, PFOS, GenX chemicals, and PFBS), 2022. https://www.epa.gov/system/files/documents/2022-06/technical-factsheet-four-PFAS.pdf (accessed December 8, 2023).

(7) Fenton, S. E.; Ducatman, A.; Boobis, A.; DeWitt, J. C.; Lau, C.; Ng, C.; Smith, J. S.; Roberts, S. M. Per- and Polyfluoroalkyl Substance Toxicity and Human Health Review: Current State of Knowledge and Strategies for Informing Future Research. *Environ. Toxicol. Chem.* **2020**, *40*, 606−630.

(8) Loccisano, A. E.; Longnecker, M. P.; Campbell, J. L., Jr; Andersen, M. E.; Clewell, H. J., 3rd Development of PBPK models for PFOA and PFOS for human pregnancy and lactation life stages. *J. Toxicol. Environ. Health, Part A* **2013**, *76* (1), 25−57.

(9) Fabrega, F.; Nadal, M.; Schuhmacher, M.; Domingo, J. L.; Kumar, V. Influence of the uncertainty in the validation of PBPK models: A case-study for PFOS and PFOA. *Regul. Toxicol. Pharmacol.* **2016**, *77*, 230−239.

(10) Perez, F.; Nadal, M.; Navarro-Ortega, A.; Fàbrega, F.; Domingo, J. L.; Barceló, D.; Farré, M. Accumulation of perfluoroalkyl substances in human tissues. *Environ. Int.* **2013**, *59*, 354−362.

(11) Abraham, K.; El-Khatib, A. H.; Schwerdtle, T.; Monien, B. H. Perfluorobutanoic acid (PFBA): No high-level accumulation in human lung and kidney tissue. *Int. J. Hyg Environ. Health* **2021**, *237*, 113830.

(12) Loccisano, A. E.; Campbell, J. L., Jr; Andersen, M. E.; Clewell, H. J., III Evaluation and prediction of pharmacokinetics of PFOA and PFOS in the monkey and human using a PBPK model. *Regul. Toxicol. Pharmacol.* **2011**, *59* (1), 157−175.

(13) East, A.; Dawson, D. E.; Brady, S.; Vallero, D. A.; Tornero-Velez, R. A Scoping Assessment of Implemented Toxicokinetic Models of Per- and Polyfluoro-Alkyl Substances, with a Focus on One-Compartment Models. *Toxics* **2023**, *11* (2), 163.

(14) Allendorf, F.; Goss, K. U.; Ulrich, N. Estimating the Equilibrium Distribution of Perfluoroalkyl Acids and 4 of Their Alternatives in Mammals. *Environ. Toxicol. Chem.* **2021**, *40* (3), 910−920.

(15) Vorkamp, K.; Nielsen, F.; Kyhl, H. B.; Husby, S.; Nielsen, L. B.; Barington, T.; Andersson, A. M.; Jensen, T. K. Polybrominated diphenyl ethers and perfluoroalkyl substances in serum of pregnant women: levels, correlations, and potential health implications. *Arch. Environ. Contam. Toxicol.* **2014**, *67* (1), 9−20.

(16) Valvi, D.; Hojlund, K.; Coull, B. A.; Nielsen, F.; Weihe, P.; Grandjean, P. Life-course Exposure to Perfluoroalkyl Substances in Relation to Markers of Glucose Homeostasis in Early Adulthood. *J. Clin. Endocrinol. Metab.* **2021**, *106* (8), 2495−2504.

(17) Baumert, B. O.; Fischer, F. C.; Nielsen, F.; Grandjean, P.; Bartell, S.; Stratakis, N.; Walker, D. I.; Valvi, D.; Kohli, R.; Inge, T.; et al. Paired Liver:Plasma PFAS Concentration Ratios from Adolescents in the Teen-LABS Study and Derivation of Empirical and Mass Balance Models to Predict and Explain Liver PFAS Accumulation. *Environ. Sci. Technol.* **2023**, *57* (40), 14817−14826.

(18) Allendorf, F.; Goss, K.-U.; Ulrich, N. Estimating the Equilibrium Distribution of Perfluoroalkyl Acids and 4 of Their Alternatives in Mammals. *Environ. Toxicol. Chem.* **2021**, *40* (3), 910−920.

(19) Allendorf, F.; Berger, U.; Goss, K. U.; Ulrich, N. Partition coefficients of four perfluoroalkyl acid alternatives between bovine serum albumin (BSA) and water in comparison to ten classical perfluoroalkyl acids. *Environ. Sci.: Processes Impacts* **2019**, *21* (11), 1852−1863.

(20) Ebert, A.; Allendorf, F.; Berger, U.; Goss, K. U.; Ulrich, N. Membrane/Water Partitioning and Permeabilities of Perfluoroalkyl Acids and Four of their Alternatives and the Effects on Toxicokinetic Behavior. *Environ. Sci. Technol.* **2020**, *54* (8), 5051−5061.

(21) Bischel, H. N.; MacManus-Spencer, L. A.; Zhang, C. J.; Luthy, R. G. Strong Associations of Short-Chain Perfluoroalkyl Acids with Serum Albumin and Investigation of Binding Mechanisms. *Environ. Toxicol. Chem.* **2011**, *30* (11), 2423−2430.

(22) Droge, S. T. J. Membrane-Water Partition Coefficients to Aid Risk Assessment of Perfluoroalkyl Anions and Alkyl Sulfates. *Environ. Sci. Technol.* **2019**, *53* (2), 760−770.

(23) Yang, D.; Han, J.; Hall, D. R.; Sun, J.; Fu, J.; Kutarna, S.; Houck, K. A.; LaLone, C. A.; Doering, J. A.; Ng, C. A.; et al. Nontarget Screening of Per- and Polyfluoroalkyl Substances Binding to Human Liver Fatty Acid Binding Protein. *Environ. Sci. Technol.* **2020**, *54* (9), 5676−5686.

(24) Jia, Y.; Zhu, Y.; Xu, D.; Feng, X.; Yu, X.; Shan, G.; Zhu, L. Insights into the Competitive Mechanisms of Per- and Polyfluoroalkyl Substances Partition in Liver and Blood. *Environ. Sci. Technol.* **2022**, *56*, 6192−6200.

(25) Khazaee, M.; Christie, E.; Cheng, W.; Michalsen, M.; Field, J.; Ng, C. Perfluoroalkyl Acid Binding with Peroxisome Proliferator-Activated Receptors α, γ, and δ, and Fatty Acid Binding Proteins by Equilibrium Dialysis with a Comparison of Methods. *Toxics* **2021**, *9* (3), 45.

(26) Zhang, L.; Ren, X.-M.; Guo, L.-H. Structure-Based Investigation on the Interaction of Perfluorinated Compounds with Human Liver Fatty Acid Binding Protein. *Environ. Sci. Technol.* **2013**, *47* (19), 11293−11301.

(27) Karrman, A.; Domingo, J. L.; Llebaria, X.; Nadal, M.; Bigas, E.; van Bavel, B.; Lindström, G. Biomonitoring perfluorinated compounds in Catalonia, Spain: concentrations and trends in human liver and milk samples. *Environ. Sci. Pollut. Res. Int.* **2010**, *17* (3), 750−758.

(28) Peters, F. T.; Steuer, A. E. Antemortem and postmortem influences on drug concentrations and metabolite patterns in postmortem specimens. *Wiley Interdiscip. Rev.: Forensic Sci.* **2019**, *1* (1), No. e1297.

(29) Michiue, T.; Ishikawa, T.; Oritani, S.; Kamikodai, Y.; Tsuda, K.; Okazaki, S.; Maeda, H. Forensic pathological evaluation of postmortem pulmonary CT high-density areas in serial autopsy cases of sudden cardiac death. *Forensic Sci. Int.* **2013**, *232* (1−3), 199−205.

(30) Shearer, J. J.; Callahan, C. L.; Calafat, A. M.; Huang, W. Y.; Jones, R. R.; Sabbisetti, V. S.; Freedman, N. D.; Sampson, J. N.; Silverman, D. T.; Purdue, M. P.; et al. Serum Concentrations of Per- and Polyfluoroalkyl Substances and Risk of Renal Cell Carcinoma. *J. Natl. Cancer Inst.* **2021**, *113* (5), 580−587.

(31) National Toxicology Program. *Monograph on Immunotoxicity Associated with Exposure to Perfluorooctanoic Acid (PFOA) and Perfluorooctane Sulfonate (PFOS)*; Research Triangle Park, NC: National Toxicology Program, 2016. https://ntp.niehs.nih.gov/sites/default/files/ntp/ohat/pfoa_pfos/pfoa_pfosmonograph_508.pdf (accessed December 8, 2023).

(32) Louisse, J.; Dellafiora, L.; van den Heuvel, J.; Rijkers, D.; Leenders, L.; Dorne, J. L. C. M.; Punt, A.; Russel, F. G. M.; Koenderink, J. B. Perfluoroalkyl substances (PFASs) are substrates of the renal human organic anion transporter 4 (OAT4). *Arch. Toxicol.* **2023**, *97* (3), 685−696.

(33) Ebert, A.; Allendorf, F.; Berger, U.; Goss, K. U.; Ulrich, N. Membrane/Water Partitioning and Permeabilities of Perfluoroalkyl Acids and Four of their Alternatives and the Effects on Toxicokinetic Behavior. *Environ. Sci. Technol.* **2020**, *54* (8), 5051−5061.

(34) Cao, H.; Zhou, Z.; Hu, Z.; Wei, C.; Li, J.; Wang, L.; Liu, G.; Zhang, J.; Wang, Y.; Wang, T.; et al. Effect of Enterohepatic Circulation on the Accumulation of Per- and Polyfluoroalkyl Substances: Evidence from Experimental and Computational Studies. *Environ. Sci. Technol.* **2022**, *56* (5), 3214−3224.

https://doi.org/10.1021/acs.est.3c06499
Environ. Sci. Technol. 2024, 58, 143−149